Words

Technical Documentation

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Repository: github.com/JuliaSivridi/Words

Stack: React 18 · JavaScript · Vite 6 · Google Sheets API v4

> A personal vocabulary learning app delivered as a serverless PWA. Words are studied through three progressively harder modes — **FlipCard → MultipleChoice → MatchingGrid** — with spaced repetition counters stored directly in Google Sheets. No backend, no subscription.

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1. Overview

Words is a personal vocabulary learning application delivered as a **Progressive Web App (PWA)**. It runs in any modern browser and installs on Android and iOS home screens as a standalone app. There is no dedicated backend server — all persistent data lives in a **Google Spreadsheet** (named `db_words`) found or created automatically via the Google Drive API v3.

Words are studied through three progressively harder exercise modes. Each mode has its own repetition counter per word. Once a word completes all three modes (24 total repetitions), it is automatically marked as learned and removed from future sessions.

Key Design Goals

- **Zero backend** — Google Sheets as a database; no server-side code
- **Any language pair** — each sheet tab is a separate language pair (e.g. `RU-EN`)
- **Spaced repetition** — three modes, each requiring progressively more repetitions
- **Category filter** — words tagged by category; sessions and word list filter by category
- **Cross-device sync** — selected language and category filter stored in the sheet
- **Mobile-first** — 16 px minimum font, touch-friendly, installable PWA
- **Dark mode** — follows OS preference; consistent color palette across all screens

2. Tech Stack

Layer	Technology	Version	Notes
UI framework	React	18	Hooks only; no class components
Language	JavaScript	ES2022	No TypeScript
Build tool	Vite	6	ES modules, HMR
Styling	CSS Modules + CSS variables	—	Light/dark via `prefers-color-scheme`
Routing	React Router DOM	6	Client-side; hash-free URLs
Database	Google Sheets API v4	—	Row-per-word; direct browser fetch
File discovery	Google Drive API v3	—	Find or create `db_words` spreadsheet
Auth	Google Identity Services	—	OAuth 2.0 implicit grant; silent sign-in
Offline storage	None	—	localStorage for language/category prefs only
PWA	Manual manifest + SW	—	manifest.json + sw.js; no Workbox
Hosting	Any static host	—	Vercel, Netlify, GitHub Pages

3. Project Architecture

The app is a single-page React application. All state is managed locally in React hooks and passed top-down from `App.jsx`. There are no global state stores — the component tree is shallow enough that prop drilling is manageable.

```
src/
├── App.jsx # root router, auth state, sheet ID, language & category
├── auth.js # GIS token client, trySilentSignIn(), onAuthChange()
├── sheetsApi.js # Drive + Sheets API fetch wrappers (all API calls)
├── constants.js # M1_MAX=4 M2_MAX=8 M3_MAX=12 TOTAL_REPS=24
├── langMap.js # tab name → display label mapping
├── theme.css # CSS variables, light/dark theme, global utilities
├── screens/
│ ├── LoginScreen.jsx # Google sign-in page
│ ├── HomeScreen.jsx # Start / Language / Category / Word List buttons
│ ├── SessionScreen.jsx # 12-step session orchestration + progress bar
│ ├── WordListScreen.jsx # scrollable word list with learned toggle
│ ├── LanguageScreen.jsx # tab selection from db_words spreadsheet
│ └── CategoryScreen.jsx # multi-select category filter
├── components/
│ ├── FlipCard.jsx # Mode 1: flip card, swipe support, Learn and hide
│ ├── MultipleChoice.jsx # Mode 2: 2×2 choice grid
│ ├── MatchingGrid.jsx # Mode 3: 6-pair matching columns
│ ├── NextButton.jsx # shared "Next →" button, sticky at bottom
│ ├── CheckIcon.jsx # shared Lucide-style checkmark SVG
│ └── WordListItem.jsx # single row in word list
└── hooks/
├── useWords.js # load words, saveSessionUpdates, setLearned, resetWord
└── useSession.js # buildSession() — 12-step session builder
```

Data Flow

1. **App start** → `trySilentSignIn()` → if ok: `findOrCreateWordsFile()` → `readSettings()`
2. Settings (language + category) restored from sheet, overriding localStorage values
3. `useWords(sheetId, tab)` fetches all rows for the selected tab via Sheets API
4. `buildSession(words, categoryFilter)` plans 12 steps; mode components render each step
5. User completes a step → counter incremented in memory (`pendingUpdates` ref)
6. After step 12 → `batchUpdateWords()` sends all counter updates to Sheets in one call
7. "Learn and hide" → immediate single-row update via `markLearned()`

4. Data Model

All data lives in the user's Google Spreadsheet named `db_words`. The app finds it in Google Drive by name and creates it automatically on first login. Each language pair is a separate sheet tab.

Word Tab (e.g. `RU-EN`)

Tab name format: `TRANSLATION-STUDY`. For `RU-EN` the user studies English; column C holds the Russian translation. Row 1 is treated as a header only if `B1 = "word"` (case-insensitive); otherwise all rows are data.

Col	Field	Type	Notes
A	category	string	Optional grouping label (e.g. "Verbs", "Food")
B	word	string	The word being studied (target language)
C	translation	string	Native language meaning
D	m1	number	FlipCard repetition count — graduates at 4
E	m2	number	MultipleChoice repetition count — graduates at 8
F	m3	number	MatchingGrid repetition count — auto-learned at 12
G	learned	`TRUE | FALSE`	Hides word from future sessions

`_settings` Tab

Stores cross-device preferences. Created automatically on first settings write.

Cell	Content
`A1`	Last selected language tab code (e.g. `RU-EN`)
`A2`	Category filter — comma-separated names, or empty string for "all"

> **Sync strategy:** Language and category are stored in both localStorage (for fast initial load) and the `_settings` tab (for cross-device sync). On login, sheet values win and overwrite localStorage.

Repetition Constants (`constants.js`)

Constant	Value	Meaning
`M1_MAX`	4	Reps needed to graduate from FlipCard mode
`M2_MAX`	8	Reps needed to graduate from MultipleChoice mode
`M3_MAX`	12	Reps needed to auto-mark word as learned
`TOTAL_REPS`	24	Total repetitions per word (4 + 8 + 12)

5. Session Algorithm

Each session consists of exactly 12 steps, built by `buildSession(words, categoryFilter)` in `src/hooks/useSession.js`. Counters are accumulated in memory and batch-saved after the final step.

Pool Building

Three word pools are built from active (non-learned) words that match the category filter:

Pool	Eligibility condition
mode1Pool	`m1 < 4`
mode2Pool	`m1 ≥ 4` AND `m2 < 8`
mode3Pool	`m2 ≥ 8` AND `m3 < 12` AND at least 6 words

Step Planning

Available modes receive equal shares of the 12 steps. Unavailable modes donate their slots evenly to the others. The resulting plan is shuffled so modes are interleaved:

Available modes	Step distribution
All three (1, 2, 3)	4 + 4 + 4
Two modes (any)	6 + 6
One mode only	12

Word Selection per Mode

- **Mode 1 & 2** — Shuffled queue: all words in the pool are cycled before any word repeats. Prevents the same word appearing multiple times in one session.
- **Mode 3** — Pool sorted by `m3` ascending, then sliced into non-overlapping groups of 6. Each mode 3 step uses the next group (`m3i % numGroups`), so different words appear on each occurrence.

Cross-Category Supplements (`buildMode3Pool`)

If a selected category has no words that qualify for mode 3 yet (i.e. no words with `m2 ≥ 8`), the app supplements the pool with the most-advanced unlearned words from that category. This ensures every

selected category is represented in the matching grid even when the user is still early in their learning progression.

6. Screens & Navigation

All routing is handled by React Router DOM v6 (`BrowserRouter`). Routes: `/`, `/session`, `/words`, `/language`, `/category`.

LoginScreen

- Horizontal logo row: app icon (icon-192.png) + "Words" heading + tagline
- Two description lines explaining Google Sheets storage and offline capability
- Orange "Sign in with Google" button — triggers GIS popup flow

HomeScreen

- Top bar: app name + user avatar (tapping signs out)
- Four buttons in order: **Start** (primary CTA) → Language → Category → Word List
- Language and Category buttons show a sub-label (selected value or "All categories")
- All buttons use monochrome Lucide-style SVG icons (`currentColor`)

SessionScreen

- Top bar: back (✕) button + step counter ("3 / 12") + thin progress bar
- Renders FlipCard, MultipleChoice, or MatchingGrid depending on `step.mode`
- NextButton always visible at the bottom (sticky via `margin-top: auto` in flex column)
- After step 12: calls `batchUpdateWords()`, shows "Session complete!" with checkmark

FlipCard (Mode 1)

- CSS 3D flip on X axis (`rotateX(180deg)`) with 0.5 s cubic-bezier transition
- Front: word + "tap to flip" hint. Back: translation on surface-colored background
- Triggers: click / tap / vertical swipe (delta Y > 50 px via `touchstart` / `touchend`)
- "Learn and hide" button: marks word as learned immediately, unlocks Next →
- Next → unlocks after first flip OR after "Learn and hide"

MultipleChoice (Mode 2)

- Top half: "Learn and hide" button + large word card
- Bottom half: 2×2 grid of translation choice cards

- Wrong tap: red flash animation (`flash-wrong` keyframe), question stays active
- Correct tap: card turns green, Next → unlocks
- Wrong answers drawn from the same category-filtered pool; falls back to full word list if < 3 options

MatchingGrid (Mode 3)

- 6 word–translation pairs; left column = shuffled words, right column = shuffled translations
- Bidirectional tap: either column can be tapped first to select; tapping opposite column attempts match
- Correct pair: both cards turn green + permanently deactivate
- Wrong pair: both flash red, selection resets
- All 6 pairs matched → Next → unlocks. No "Learn and hide" in this mode

LanguageScreen

- Lists all sheet tabs from `db_words` (excludes `Sheet1`, `Лист1`, `_settings`)
- Selected tab shown with orange checkmark on right (shared `<CheckIcon>` component)
- Tapping a tab saves to localStorage + `_settings` and navigates back to Home

CategoryScreen

- Multi-select: any combination of categories can be active simultaneously
- "All categories" row at top — toggles between full selection and empty selection
- Back button saves selection and navigates to Home (no separate "Done" button)
- Stored as `null` (all) or `string[]`; synced to sheet as comma-separated string

WordListScreen

- Header shows "learned / total" counter for visible words
- Each row: word | translation | progress ("3 / 24") | toggle icon
- Eye icon: tap to mark as learned (saved immediately). Eye-crossed: tap to reset all counters
- Filtered by `sessionCategory` if set

7. Authentication

Authentication uses **Google Identity Services (GIS)** with the OAuth 2.0 implicit token grant. No redirect URIs are needed — only Authorized JavaScript Origins must be configured.

- On app load: `trySilentSignIn()` calls `tokenClient.requestAccessToken({ prompt: '' })`
- Empty `prompt` = no UI if user previously granted access; popup only on first login
- User profile (`name`, `email`, `picture`) fetched from Google userinfo endpoint after token received
- Email saved in localStorage as `login_hint` for subsequent silent sign-ins
- Access token stored in module-level memory variable only (never persisted)
- On 401 from Sheets API: silent token refresh attempted automatically
- Sign-out: clears user object and calls `google.accounts.oauth2.revoke()`

Required OAuth Scopes

- `email`, `profile`
- `https://www.googleapis.com/auth/spreadsheets`
- `https://www.googleapis.com/auth/drive.metadata.readonly`

> **Google Cloud Console setup:** Enable Google Sheets API v4 *and* Google Drive API. Create an OAuth 2.0 Client ID (Web application type). Add your domain to *Authorized JavaScript Origins* (not Redirect URIs). Add your Google account as a test user in the OAuth consent screen.

8. Google Sheets API Integration

All API calls are plain `fetch()` requests in `src/sheetsApi.js`, using the in-memory access token. A central `request(url, options)` helper attaches the Bearer header and calls `refreshTokenIfNeeded()` before every request.

Key API Calls

Operation	Endpoint
Find spreadsheet	`GET drive/v3/files?q=name='db_words'&mimeType=...spreadsheet...`
Create spreadsheet	`POST sheets/v4/spreadsheets` with `{ title: 'db_words' }`
List tabs	`GET sheets/v4/spreadsheets/{id}?fields=sheets.properties.title`
Read words	`GET sheets/v4/spreadsheets/{id}/values/{tab}!A1:G`
Read settings	`GET sheets/v4/spreadsheets/{id}/values/_settings!A1:A2`
Batch update counters	`POST sheets/v4/spreadsheets/{id}/values:batchUpdate` — range `{tab}!D{row}:G{row}`
Mark learned	`PUT sheets/v4/spreadsheets/{id}/values/{tab}!G{row}`
Reset word	`PUT sheets/v4/spreadsheets/{id}/values/{tab}!D{row}:G{row}` — `[[0, 0, 0, "FALSE"]]`
Write settings	`PUT sheets/v4/spreadsheets/{id}/values/_settings!A1:A2`

Batch Save Strategy

- Counter increments are accumulated in a `pendingUpdates` ref during the session
- `batchUpdateWords()` is called once after step 12 — one API call for all 12 updates
- "Learn and hide" triggers an immediate single-row update (`markLearned()`)
- `resetWord()` writes `[[0, 0, 0, "FALSE"]]` to columns D–G to clear all counters
- Sheet ID cached in localStorage under `words_sheet_id` to avoid Drive API calls on repeat visits

9. UI & Theme

Color System

All colors are CSS variables defined in `src/theme.css`. Dark mode switches automatically via `@media (prefers-color-scheme: dark)`.

Variable	Light	Dark	Usage
`--bg`	#FFFFFF	#1C1C1C	App background
`--surface`	#F3F1EF	#363636	Headers, secondary backgrounds
`--surface-2`	#E8E8E8	#424242	Hover states
`--text`	#1C1C1C	#EFEFEF	Primary text
`--text-muted`	#6B6B6B	#949494	Secondary text, counters
`--accent`	#E07E38	#E8935A	Buttons, checkmarks, progress bar
`--accent-hover`	#C96E2F	#D4804A	Hover on accent elements
`--success`	#6BBF7A	#5AA569	Correct answers, learned state
`--error`	#E07070	#C96060	Wrong answers
`--border`	#E0E0E0	#4A4A4A	Card borders, dividers
`--card-bg`	#FFFFFF	#363636	Flashcard backgrounds

> **Consistency:** The accent color (#E07E38) is intentionally identical to the primary color in the companion Tasks project (`HSL 25 75% 55%`). In dark mode the accent is lightened to #E8935A for better contrast on dark backgrounds.

Icons

All icons are inline SVGs in the Lucide style: `viewBox="0 0 24 24"`, `stroke="currentColor"`, `strokeWidth="1.75"`, `strokeLinecap="round"`, `strokeLinejoin="round"`. A shared `<CheckIcon size={n}>` component is used for all checkmark indicators (language selection, category selection, word list learned state, session complete screen). No icon library package is installed — paths are inlined directly.

Touch & Mobile

- Minimum tap target: 44 px on all interactive elements
- `-webkit-tap-highlight-color: transparent` on all buttons
- FlipCard: vertical swipe (delta Y > 50 px) triggers flip
- All screens: `height: 100dvh; overflow: hidden` — no page-level scroll
- Content that may exceed viewport (word list) uses internal `overflow-y: auto`

10. PWA & Mobile

- Web App Manifest (`public/manifest.json`): name "Words", display `standalone`
- Theme color: `#E07E38` in both `manifest.json` and `<meta name="theme-color">`
- Icons: `icon-192.png` and `icon-512.png` — orange rounded square with white "W"
- Service worker (`public/sw.js`): App Shell caching for HTML, JS, CSS, icons
- GIS script (`accounts.google.com/gsi/client`) loaded with `async defer` in `index.html`
- Android: Chrome auto-prompts for installation or browser menu → Install app
- iOS: Safari → Share button → Add to Home Screen

11. Setup & Deployment

Prerequisites

- Google account
- Google Cloud project with **Google Sheets API v4** and **Google Drive API** enabled
- OAuth 2.0 Client ID (type: Web application)
- Node.js ≥ 18

Google Cloud Console

1. Enable **Google Sheets API v4** and **Google Drive API**
2. Create an **OAuth 2.0 Client ID** → Web application
3. Add to **Authorized JavaScript Origins** (not Redirect URIs):

```
http://localhost:5173
https://your-app.vercel.app
```

4. Add your account as a **test user** in the OAuth consent screen

Prepare Your Google Sheet

The app automatically creates the `db_words` spreadsheet on first login. You only need to add a tab and populate it:

1. Open the created `db_words` spreadsheet in Google Drive
2. Add a tab named like `RU-EN` (format: `TRANSLATION-STUDY`)
3. Fill columns: **A** = category (optional), **B** = word, **C** = translation
4. Row 1 can be a header or data — the app auto-detects based on whether B1 = "word"

Local Development

```
git clone https://github.com/JuliaSivridi/Words.git
cd Words
npm install
```

Create `.env` in the project root:

```
VITE_GOOGLE_CLIENT_ID=your-client-id.apps.googleusercontent.com
```

```
npm run dev    # http://localhost:5173
npm run build  # production build → dist/
```

Deploy to Vercel

1. Import the repository at **vercel.com**
2. Add environment variable in Settings → Environment Variables: `VITE_GOOGLE_CLIENT_ID`
3. Every push to `main` triggers automatic deployment
4. Add the Vercel URL to Google Cloud Console Authorized JavaScript Origins

> **Single environment variable:** Unlike the companion Tasks project, Words requires only `VITE_GOOGLE_CLIENT_ID`. The spreadsheet ID is discovered automatically via the Drive API and cached in localStorage — no `VITE_SPREADSHEET_ID` needed.